



NETSEC HUB

Professional Networking & Cybersecurity Training

Cisco Certified Network Professional (CCNP)- Enterprise – Advanced Routing & Networking

- ***Mode: Online***
- ***Duration: 3-3.5 Months***

Course Overview:

CCNP Enterprise Industry Skill Development Program:

This course is designed to develop advanced enterprise networking skills required for Network Engineer and Network Security Engineer roles. The focus of this program is not limited to certification exam preparation; instead, it focuses on advanced routing concepts, enterprise network design, configuration, troubleshooting, and real-world implementation approaches.

Students will learn how enterprise networks are designed, optimized, and maintained using advanced routing technologies followed in industry environments. The course includes live demonstrations, practical configuration sessions, troubleshooting scenarios, Q&A discussions, and interview preparation to help students develop job-oriented networking skills.

Hands-on concepts will be demonstrated using network simulation environments such as EVE-NG. Students will learn practical implementation of advanced routing technologies according to the course topics. *(Lab files and personal lab infrastructure are not included with the course.)*

Course Modules:

Module 1: Advanced EIGRP

- EIGRP Architecture, Operation and Packet Types
 - Neighbor Formation, Adjacency and Troubleshooting
 - DUAL Algorithm, Metric Calculation, K-Values, Administrative Distance and Feasible Distance
 - Successor, Feasible Successor and Feasibility Condition
 - Variance and Unequal Cost Load Balancing
 - EIGRP Stub, Passive Interface, Authentication and Summarization
 - EIGRP Troubleshooting and Optimization
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Module 2: Advanced OSPF

- OSPF Architecture, Operation and Packet Types
 - Neighbor Formation, OSPF States and Troubleshooting
 - OSPF Tables, LSAs and SPF Algorithm
 - OSPF Network Types and DR/BDR Election
 - OSPF Areas and Router Types
 - Stub Area, Totally Stub Area, NSSA and Totally NSSA
 - Virtual Links, Authentication and Summarization
 - OSPF Troubleshooting and Optimization
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Module 3: Border Gateway Protocol (BGP)

- BGP Fundamentals, Autonomous Systems and BGP Operation
 - eBGP, iBGP and Neighbor Peering (Physical, Loopback and Multihop Peering)
 - BGP States, Packet Types and Troubleshooting
 - BGP Full Mesh, Route Reflectors and Next-Hop-Self
 - BGP Peer Groups and Route Management
 - BGP Path Selection and Attributes (Weight, Local Preference, AS Path, MED, Origin)
 - BGP Communities, Route Tagging, Confederation and Route Filtering
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Module 4: Enterprise Routing Services & WAN Technologies

- Route Redistribution and Route Filtering
- Policy Based Routing (PBR)
- VRF and VRF Lite Concepts
- MPLS Fundamentals, Labels, LDP and Forwarding Process
- MPLS Layer 3 VPN, VRF and MP-BGP Concepts
- Multi-Customer MPLS Network Implementation
- DMVPN Architecture, GRE Tunnel, NHRP and Hub-Spoke Design
- DMVPN Phase 1, Phase 2 and Phase 3 Overview
- Enterprise WAN Design and Troubleshooting

Career Outcome

After completing this program, students will develop advanced enterprise routing skills along with practical configuration and troubleshooting knowledge required for intermediate networking roles such as:

- Network Engineer (L2)
- Enterprise Network Engineer
- Network Security Engineer
- Infrastructure Engineer

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